

SUURET MUINAISET TECHNICAL MANUAL
Version 1.0

Installation: Suuret Muinaiset
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GitHub: https://github.com/antarthureau/suuret_muinaiset

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ELECTRICAL SAFETY WARNING

- **ALWAYS disconnect 220V AC power before opening any enclosure**
- **ONLY power units with open enclosures when necessary safety precautions are taken**
- **Use proper lockout/tagout procedures during maintenance**
- **Ant Art Enkeltpersonforetak is not liable for accidents due to misuse of the equipment or misinterpretation of these instructions**

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QUICK START FOR TECHNICIANS

Closed enclosure:

1. Connect USB cable to monitor serial output to communicate with units

Open enclosure (powered on):

1. Verify 12V and 36V PSU indicator LEDs are illuminated, test voltages them with multimeter (230V AC, 12V DC, 36V DC)
2. Verify continuity on relays and all screw terminals
3. Refer to system status or error codes using the 4-LEDs array
4. See troubleshooting flowchart

Parts that are prone to failure earlier are the audio amplifier and the relays. If one channel of the amplifier fails, use the second channel. If both channels fail, replace amplifier board. There are only 2 relay in used on the 4 relays board, if one relay fail, use another one and re-cable properly between the motherboard and the relay board. If all fail, replace the relay board.

SYSTEM OVERVIEW

This system consists of 3 units playing sound and controlling light for the installation Suuret Muinaiset. In the installation, 3 concrete sculptures/creatures communicate through sound and light:

- LONG (Green) - Leader unit (ID0)
- SMALL (Yellow) - Follower unit (ID1)
- SEASHELL (Red) - Follower unit (ID2)

Each unit consists of:

- IP65-rated enclosure containing:
 - Teensy 4.0 microcontroller
 - Custom motherboard with I/O connections
 - Official Teensy Audio Shield
 - Dual-channel 150W audio amplifier
 - 36V DC PSU (audio power)
 - 12V DC PSU (control/LED power)
- Outdoor-rated speaker (Dayton WP8BT)
- LED strip mounted on speaker grille

Programming Environment: Arduino + Teensyduino frameworks with external libraries

Communication Architecture

The 3 units communicate via Software Serial3 (TTL signal over underground CAT6 cable). The implementation uses a leader-follower pattern:

- Leader unit (LONG) sends synchronized commands to both followers
- Followers can report status back to leader when requested
- Each unit can be independently controlled via USB serial connection
- All units run identical firmware; behavior determined by hardware ID pins

Automated Operation

The leader unit includes an RTC module ensuring:

- Puts system to sleep at 22:00 (36V DC off during that time)
- Wakes system at 07:00 (36V DC on during that time)
- Forwards sleep/wake commands to follower units

Diagnostics

Each motherboard features a 4-LED binary array displaying values 0-16 for system status and troubleshooting.

NORMAL OPERATION INDICATORS

- LED Array: Binary 1 (0001) = Normal operation
- Audio: periodic sound sequence as programmed (loops every 6mn during wakeup)
- Light: periodic light sequence as programmed (loops every 6mn during wakeup)
- Power: Both PSU indicator LEDs illuminated during wakeup, only 12V DC during sleep.
- Sleep Schedule: 22:00 - 07:00 (adjustable via reprogramming)
- Serial Output: Regular status messages, no errors

REQUIRED TOOLS

Basic Maintenance:

- Flat screwdriver 3mm - terminal blocks
- Flat screwdriver 6mm - enclosure fasteners
- Phillips screwdriver 6mm - general use
- Multimeter - voltage/continuity testing

Board-Level Service:

- Phillips 000 or 00 - PCB work
- Hex key 4mm - motherboard mounting
- Hex key 4.5mm - module mounting
- Anti-static wrist strap

On-site testing Equipment:

- Wago connectors - temporary connections
- Crocodile clips - testing leads
- 3.5mm stereo headphones - audio testing
- Computer with:
 - USB-A port
 - Arduino IDE + Teensyduino installed
 - Serial monitor software (PuTTY, Arduino Serial Monitor, etc.)

Advanced Diagnostics: (troubleshooting in workshop environment)

- Arduino UNO R3 - RTC module reprogramming and logic testing/analysis
- Oscilloscope - signal analysis
- Spare Teensy 4.0 - component swap testing
- Bench power supply - isolated testing

WARRANTY AND SUPPORT

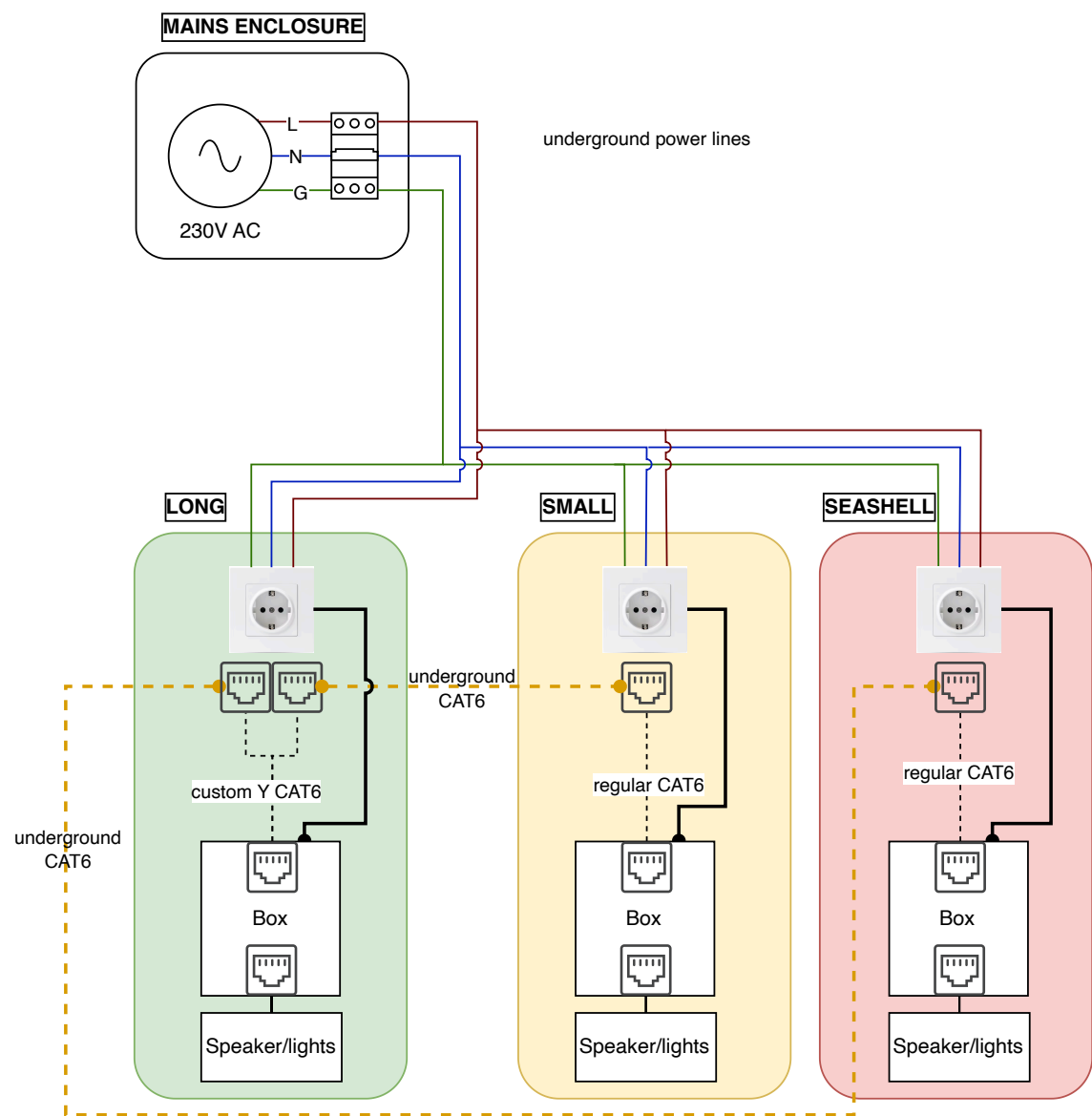
Warranty Period: 04.06.2025 - 04.06.2026

Support Contact during warranty period: ahureau@pm.me (designer) / artturi.elovirta@gmail.com (support technician in Turku)

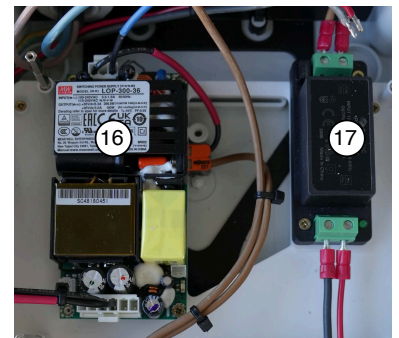
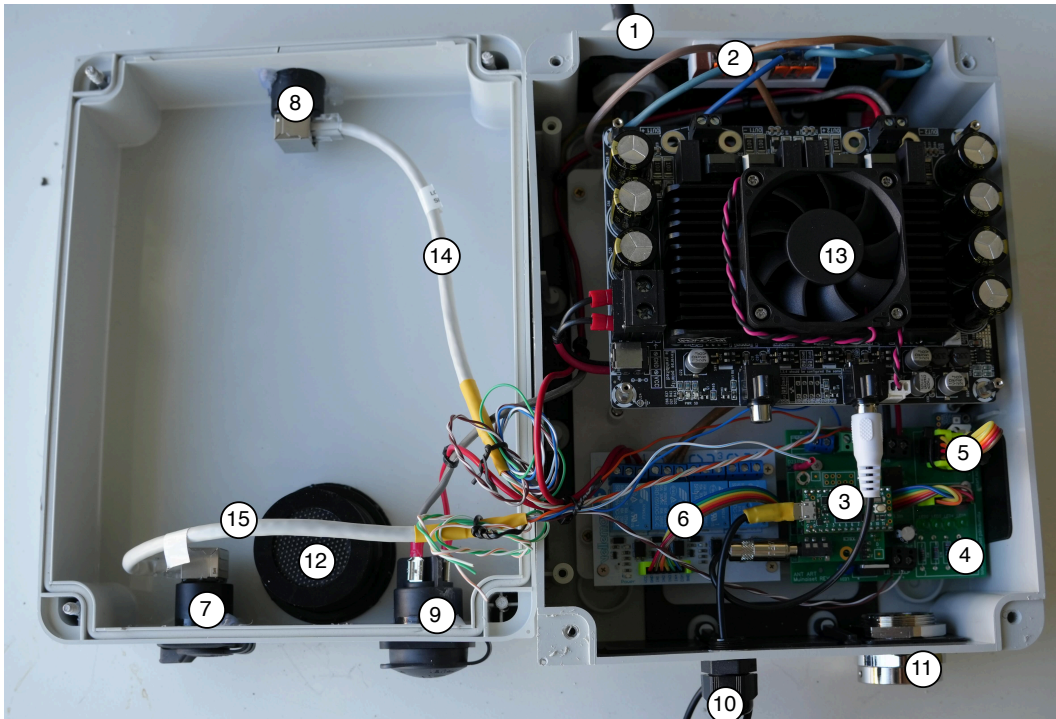
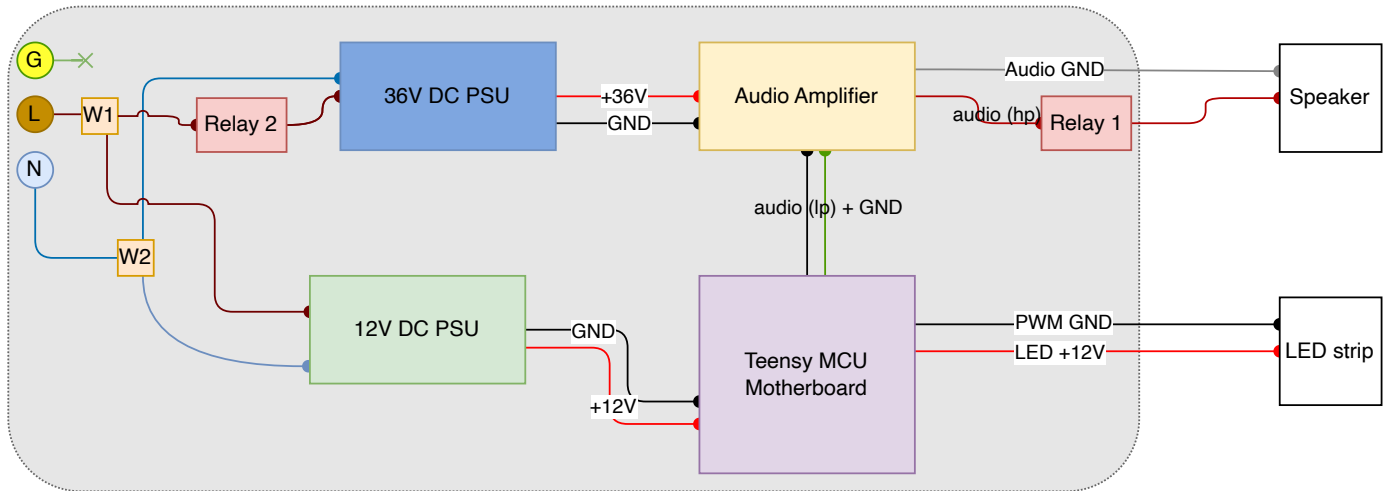
Response Time: 24-48 hours

In the following pages, the units are detailed through comprehensive diagrams, pinout references, and step-by-step maintenance procedures.

POWER AND NETWORK WIRING ON SITE



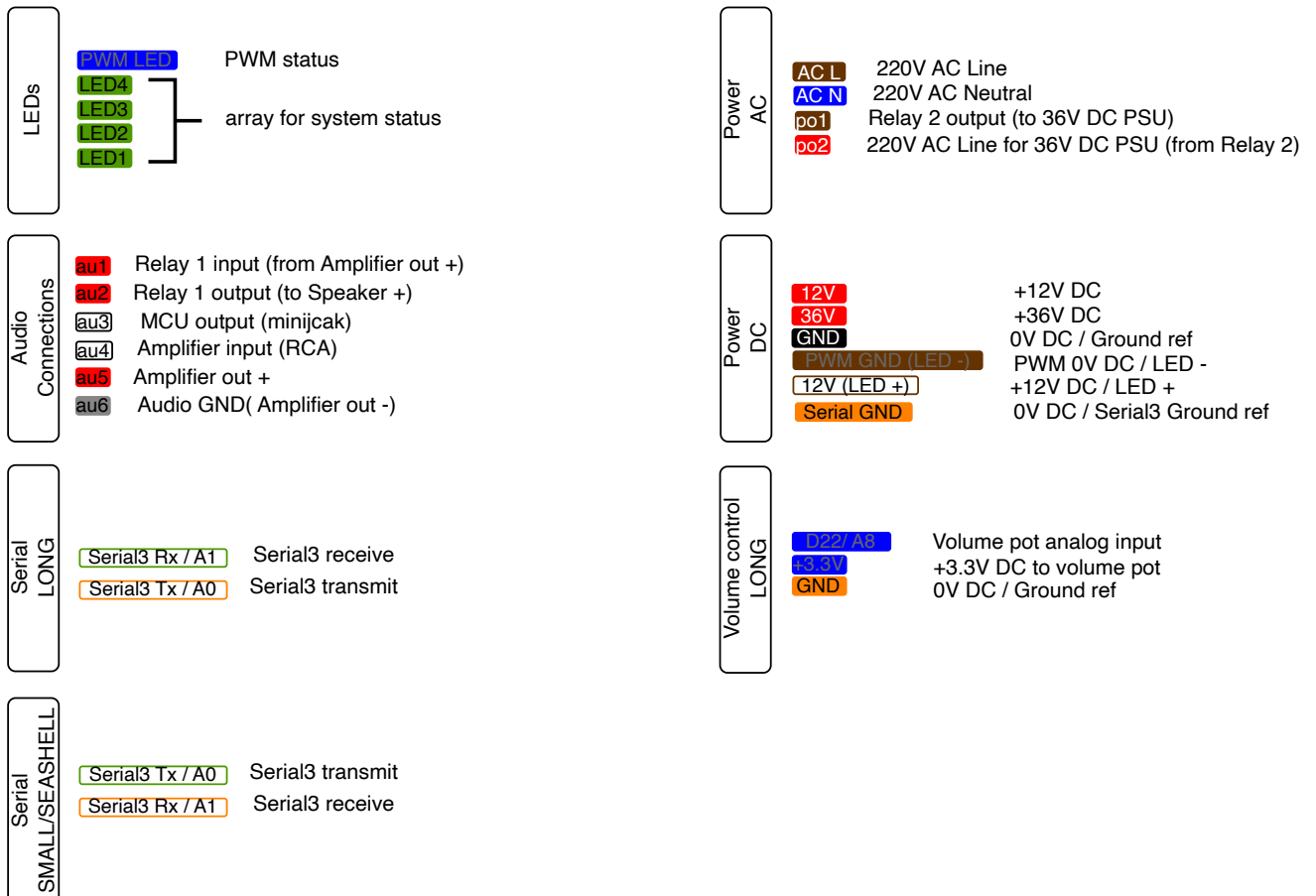
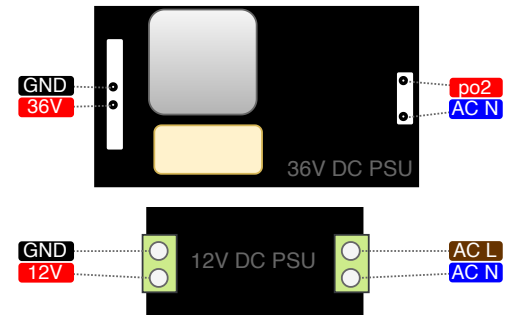
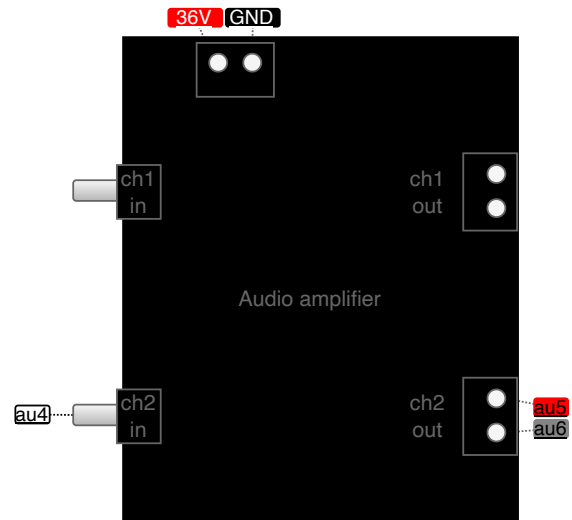
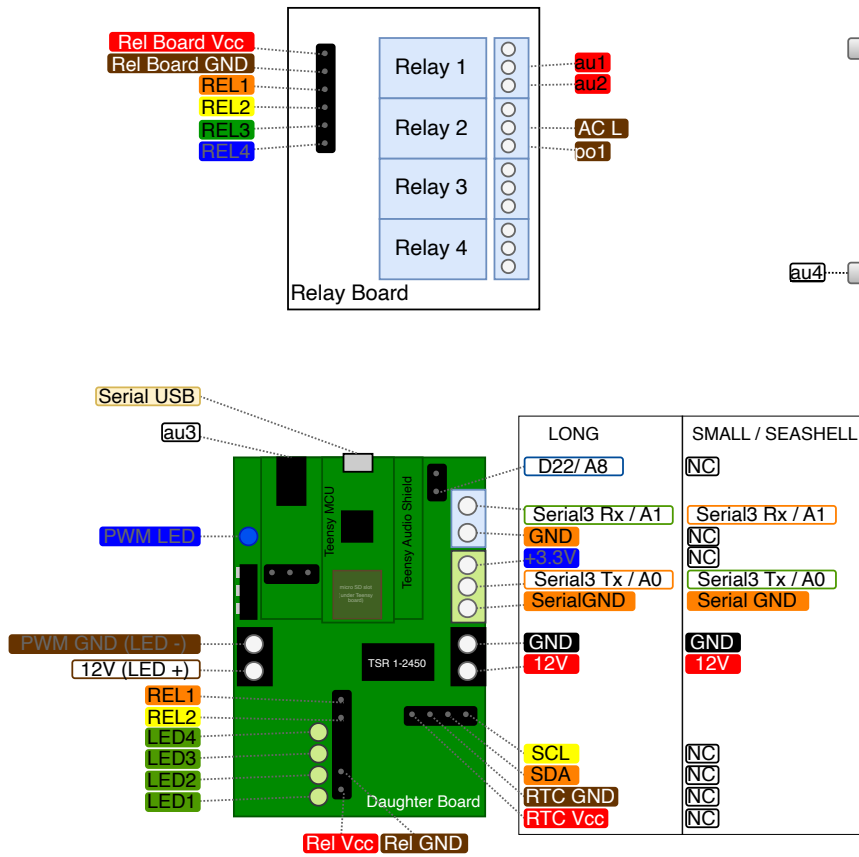
BLOCK DIAGRAM WIRING IN LONG / SMALL / SEASHELL



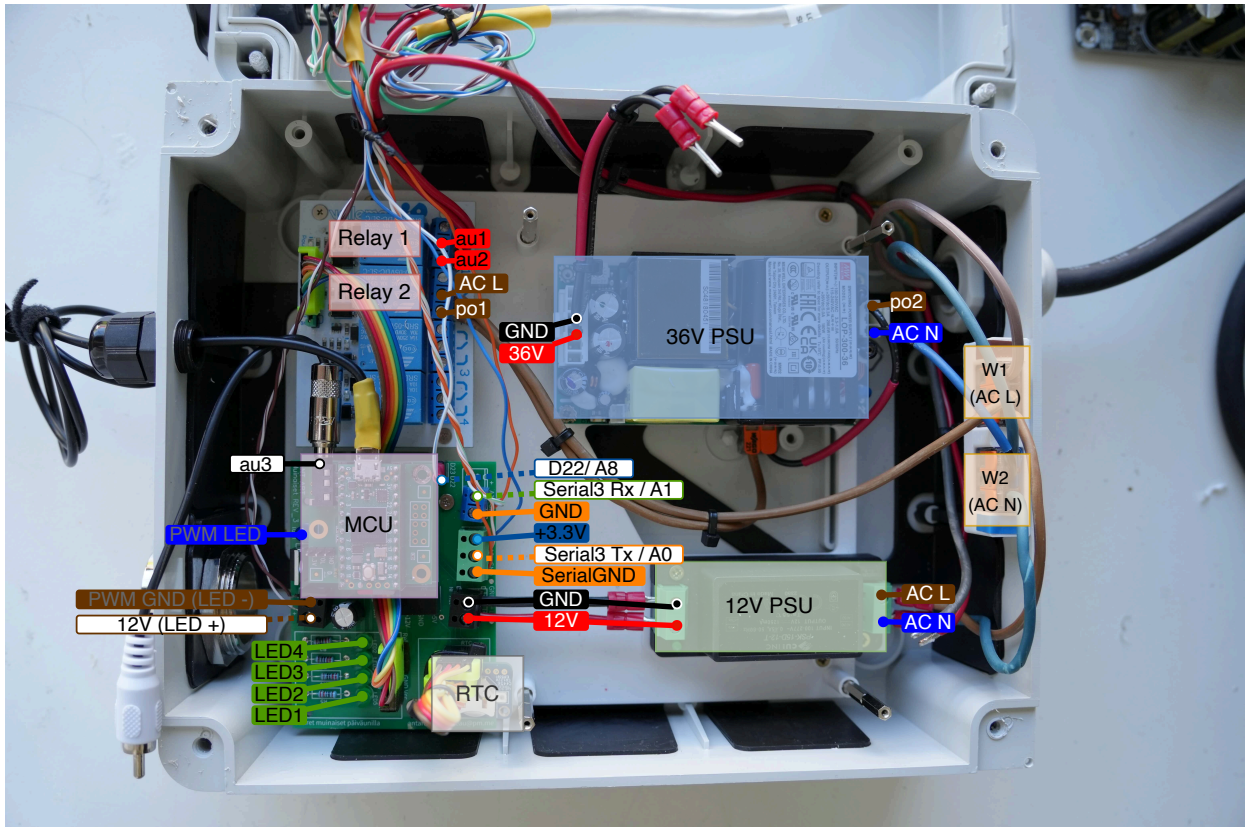
16. 220V AC to 36V DC PSU
17. 220V AC to 12V DC PSU

1. Main power input (220V AC)
2. Wagos for 220V AC distribution/splitting
3. Teensy 4.0 MCU
4. MCU Motherboard with different I/O and LED array
5. RTC module (LONG player only)
6. Relay board to switch certain connections on/off
7. EtherCON chassis connector serving as PWM output to LED strip
8. EtherCON chassis connector serving as a serial communication port
9. PowerCON chassis connector serving as audio output to speaker
10. USB Cable to upload code and communicate with the MCU
11. Pressure compensating vent to maintain stable pressure and temperature in unit
12. Silica gel tank to absorb humidity in unit
13. Dual channel 150W audio amplifier
14. CAT6 cable used to link motherboard to the chassis connector (Serial)
15. CAT6 cable used to link motherboard to the chassis connector (PWM)

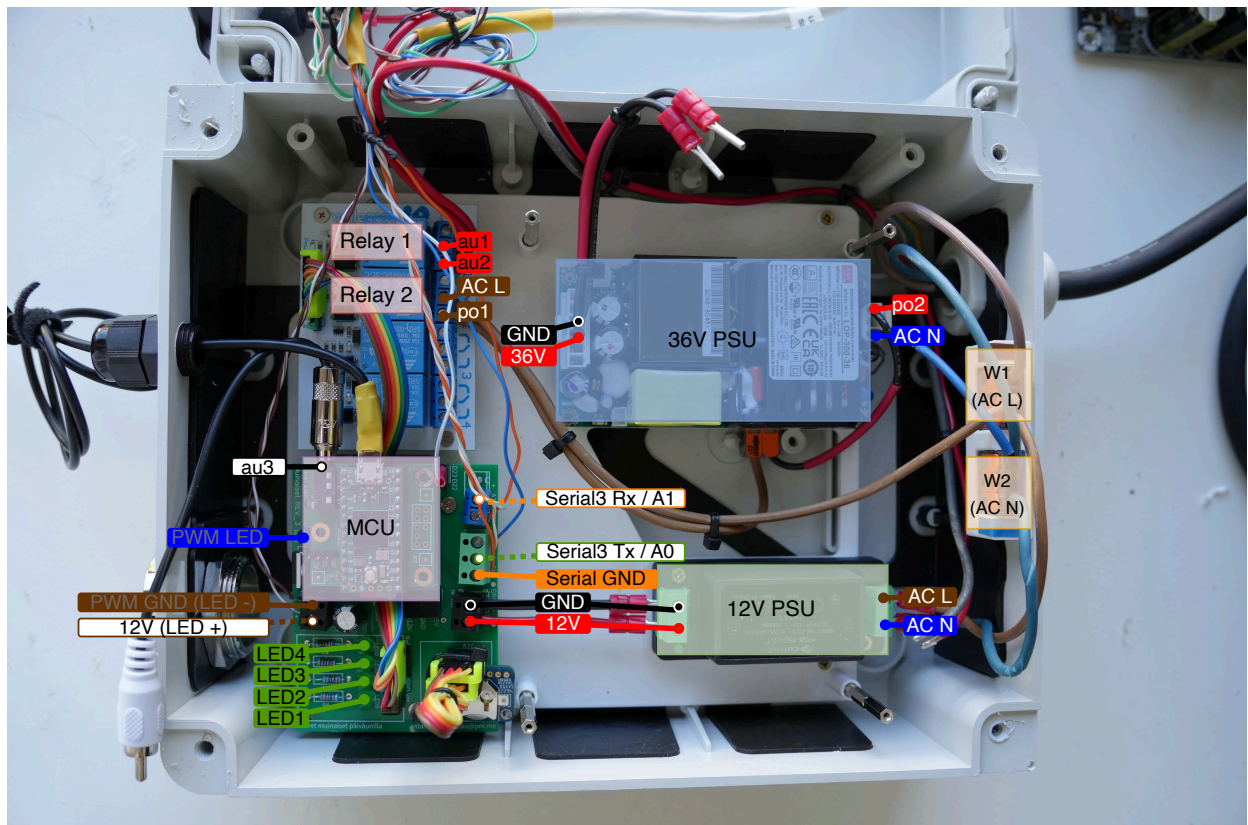
PINOUT REFERENCES CHART



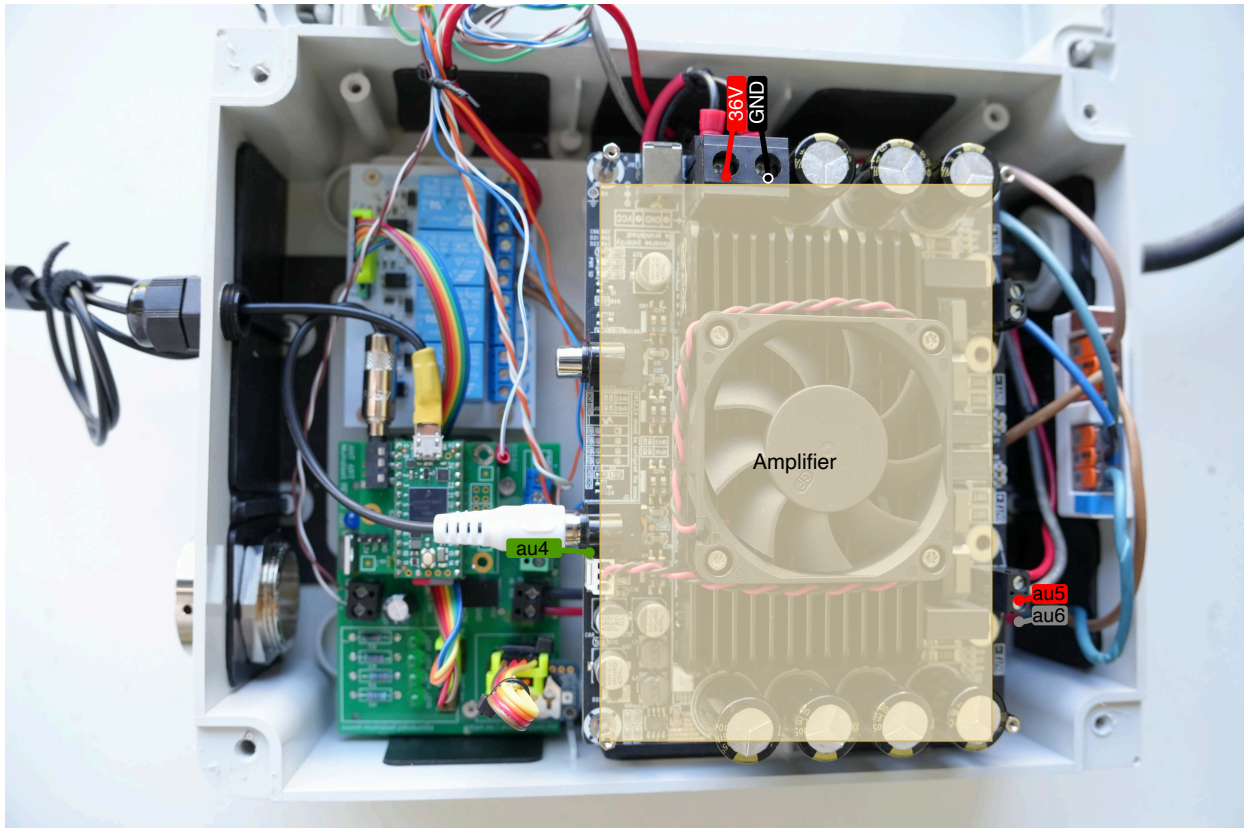
LONG PINOUT



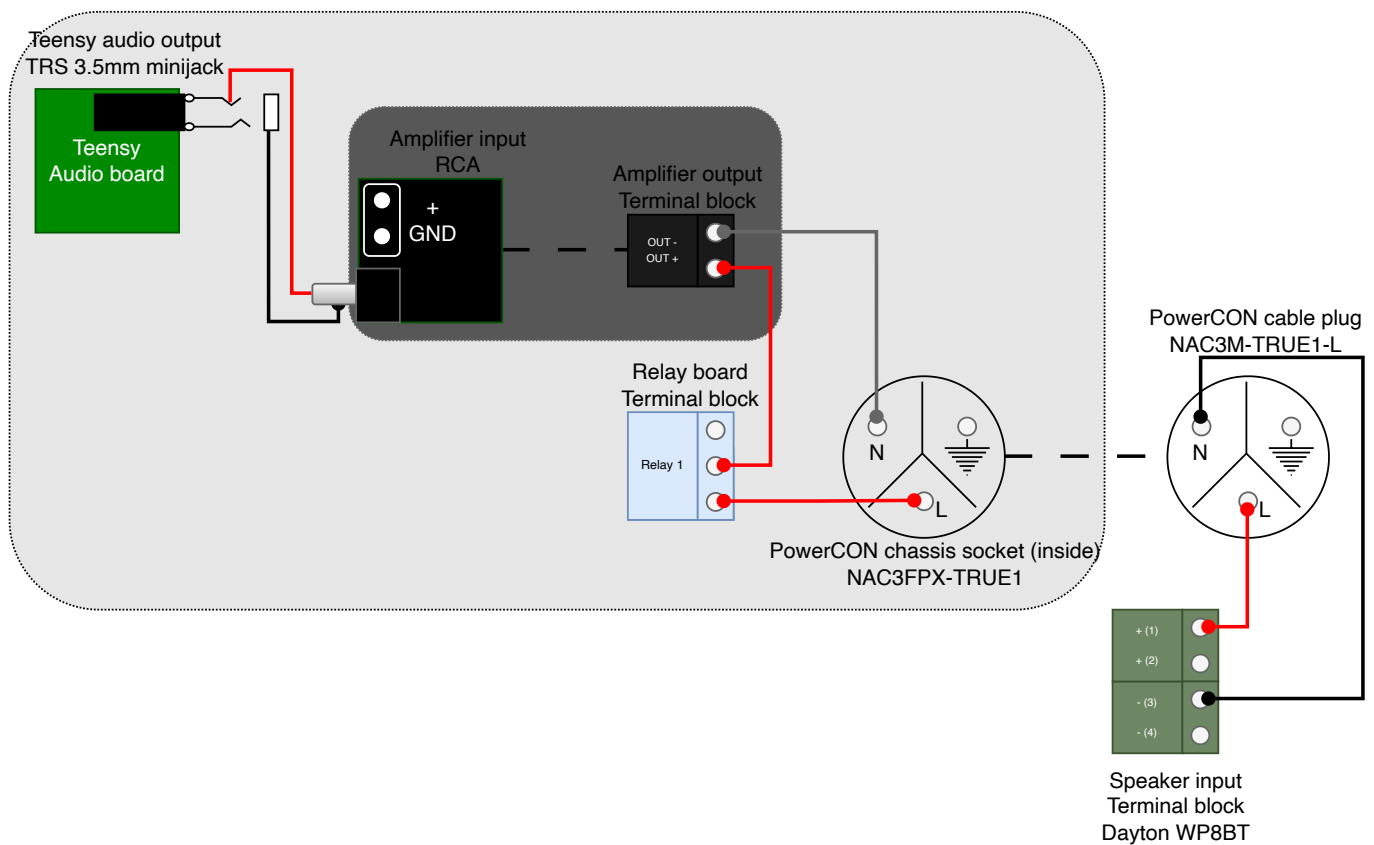
SMALL / SEASHELL PINOUT



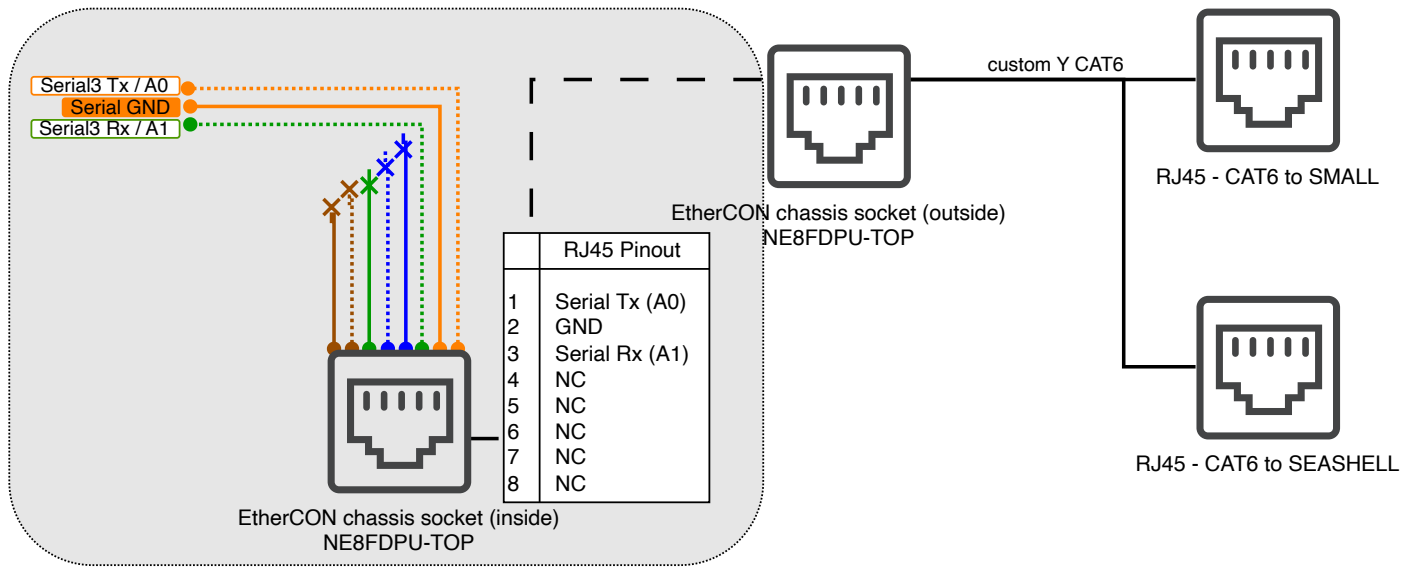
LONG/SMALL / SEASHELL AMPLIFIER PINOUT



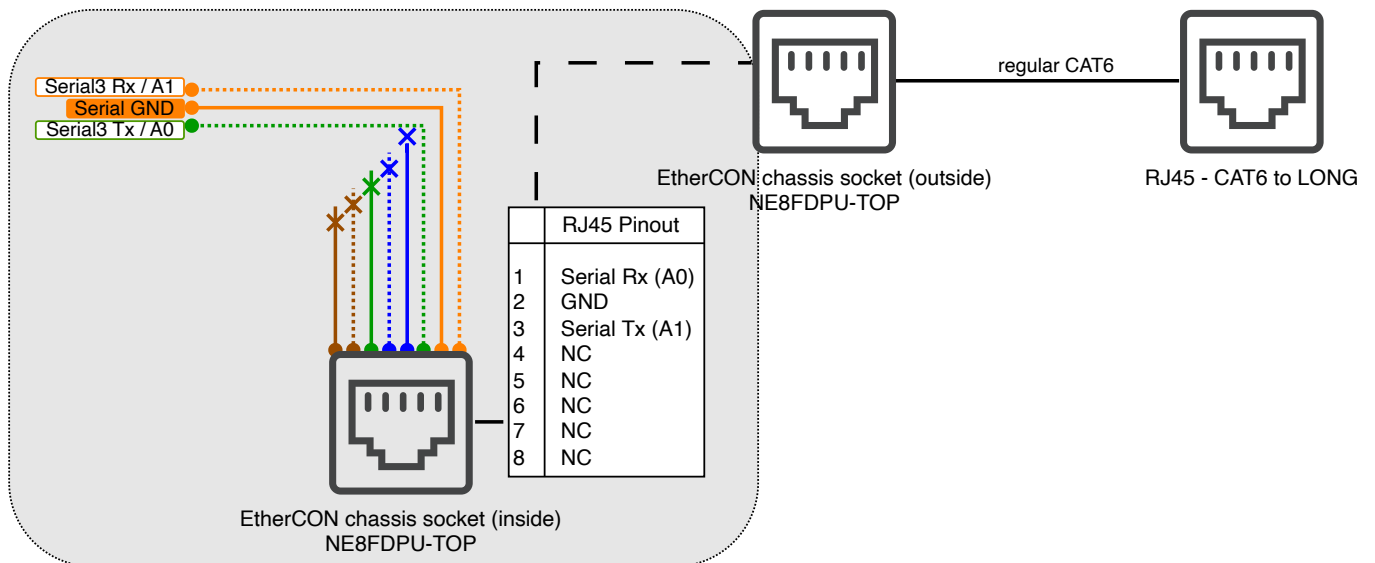
LONG / SMALL / SEASHELL - AUDIO SIGNAL CHAIN



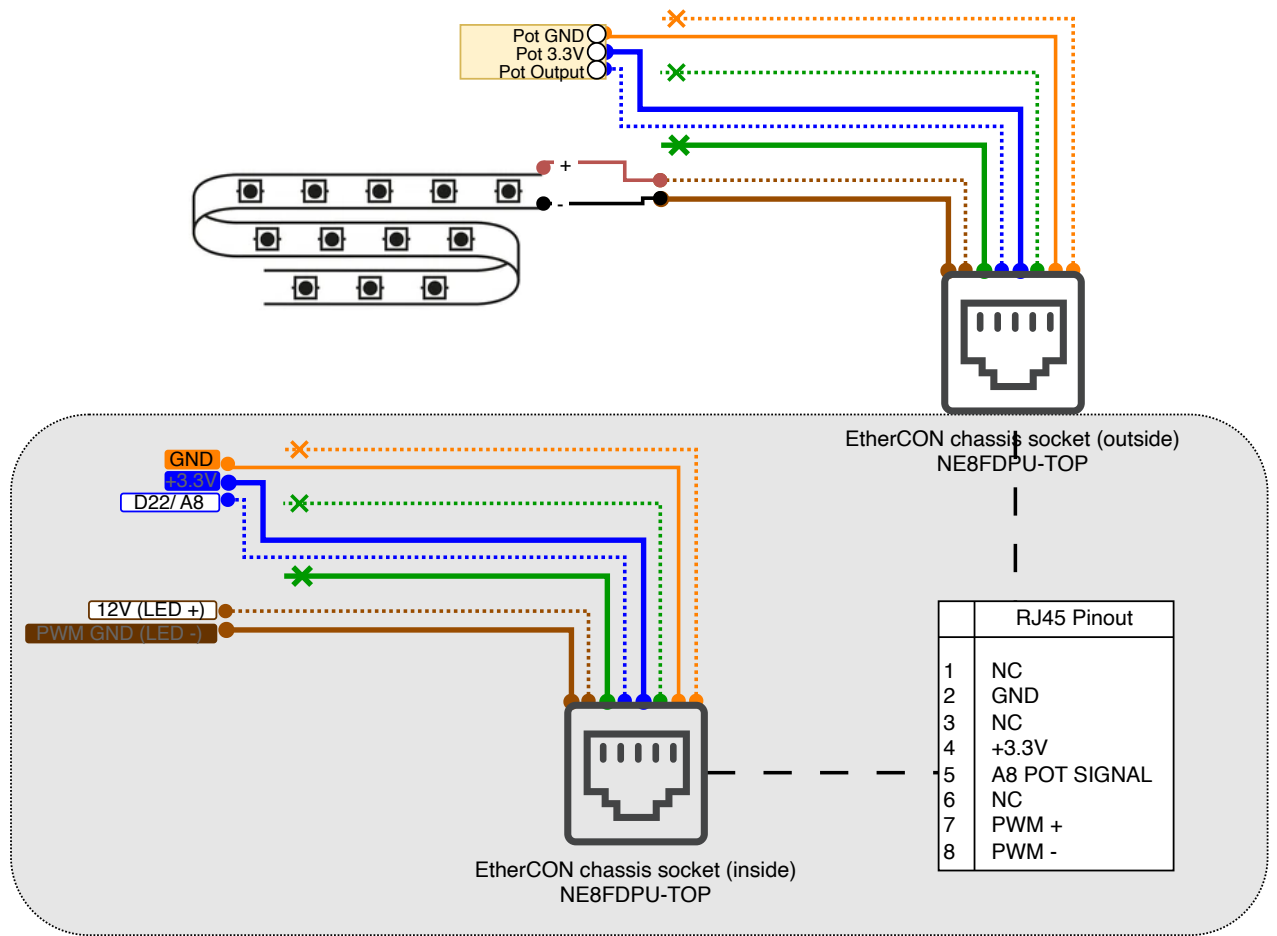
LONG - SERIAL WIRING



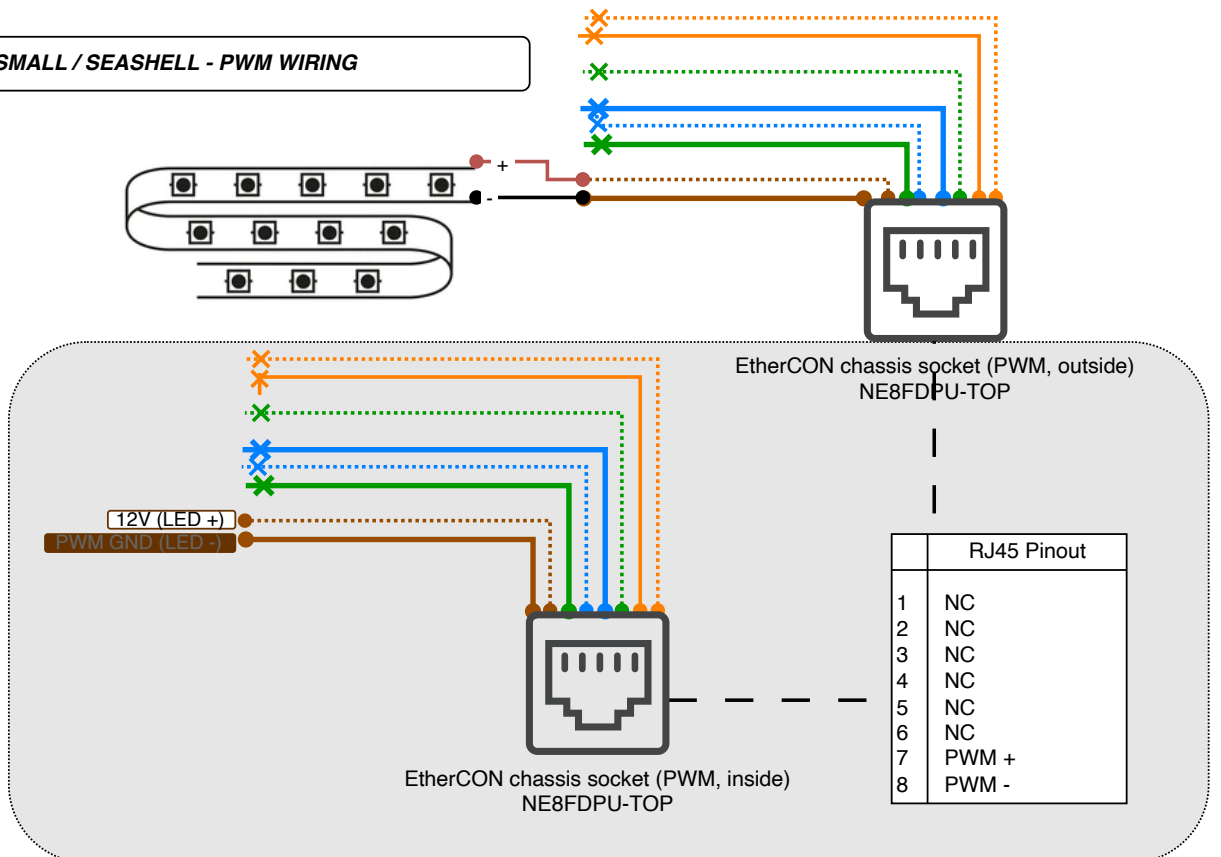
SMALL / SEASHELL - SERIAL WIRING



LONG - PWM / VOLUME CTRL WIRING

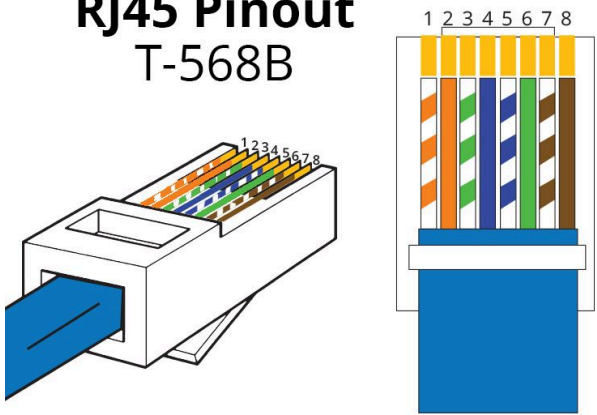


SMALL / SEASHELL - PWM WIRING



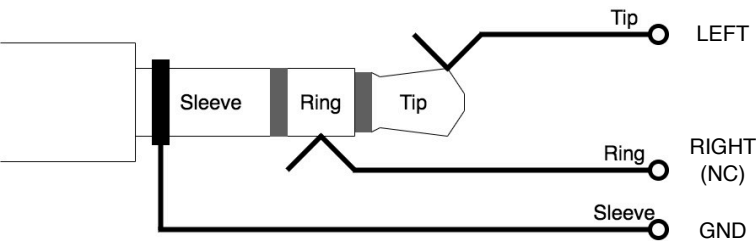
REFERENCES

RJ45 Pinout
T-568B



- | | |
|-----------------|----------------|
| 1. White Orange | 5. White Blue |
| 2. Orange | 6. Green |
| 3. White Green | 7. White Brown |
| 4. Blue | 8. Brown |

TRS 3.5 minijack pinout



outside the enclosure

inside the enclosure